



1



Mark for Review



As part of a science experiment on evaporation, Ella measured the height of water in a glass over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters (cm), of the water in the glass x days after the start of the experiment. Which of the following is the best interpretation of **39** in this context?

- (A) The estimated height, in cm, of the water at the start of the experiment
- (B) The estimated height, in cm, of the water at the end of the experiment
- (C) The estimated change in the height, in cm, of the water each day
- (D) The estimated number of days for all the water to evaporate

2.

Area (square feet)	Water (gallons)
2,520	4,536
3,780	6,804
5,040	9,072

The roofs on the buildings of a public garden are designed to allow water to drain into the garden. The table shows the relationship between the area x , in square feet, of a roof and the amount of water $f(x)$, in gallons, drained from the roof into the garden over a certain period of time. Which equation could define f ?

- A. $f(x) = 0.6x$
- B. $f(x) = 1.8x$
- C. $f(x) = 2,268x$
- D. $f(x) = 1,260x$



3



Mark for Review



In 2010, a skating team had a total of 35 skaters, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced skaters on the team increased by approximately 53%, and the number of intermediate skaters on the team increased by approximately 44%. The total number of skaters on the team increased by approximately 49%. Which equation best represents this situation, where a represents the number of advanced skaters on the team in 2010 and b represents the number of intermediate skaters on the team in 2010?

(A) $1.53a + 1.49b = 35(1.44)$

(B) $1.49a + 0.53b = 35(1.44)$

(C) $1.53a + 1.44b = 35(1.49)$

(D) $1.44a + 1.53b = 35(1.49)$

4



Mark for Review



$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

(A) $x = \frac{b-72}{y}$

(B) $x = by - 72$

(C) $x = \frac{by-72}{y}$

(D) $x = by - 72y$

5



Mark for Review



$$y = \frac{2}{7}x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b ?

(A) -3

(B) $-\frac{1}{3}$

(C) $\frac{1}{3}$

(D) 3

6



Mark for Review



$$y = 6x + 16$$

$$-7x - y = 36$$

What is the solution (x, y) to the given system of equations?

(A) $(-4, -8)$

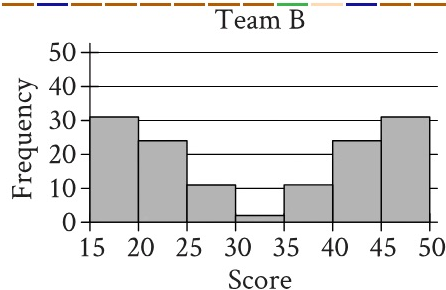
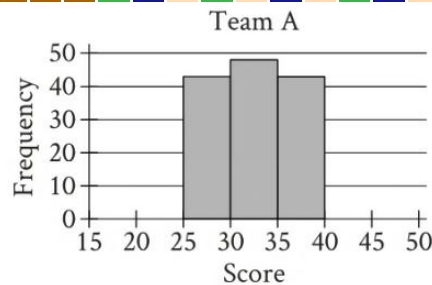
(B) $(-\frac{20}{13}, -\frac{80}{13})$

(C) $(4, 40)$

(D) $(20, 136)$



7.



The histograms summarize the distributions of scores for team A and team B. Which statement best compares the standard deviations of scores for these teams?

- (A) The standard deviation of scores for team A is less than the standard deviation of scores for team B.
- (B) The standard deviation of scores for team A is greater than the standard deviation of scores for team B.
- (C) The standard deviation of scores for team A is equal to the standard deviation of scores for team B.
- (D) There is not enough information to compare the standard deviations.

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$

8



Mark for Review

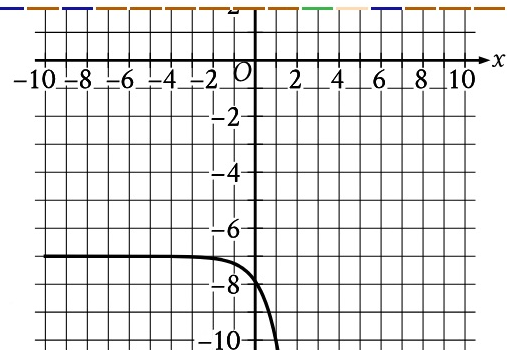
$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

Answer Preview:



9.



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a , b , and c are constants. For how many values of x does $f(x) = 0$?

(A) Three

(B) Two

(C) One

(D) Zero

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Examples

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3.5	3.5 3.50 7/2	$3\frac{1}{2}$ $3\frac{1}{2}$

10



Mark for Review

To study fluctuations in leaf water potential, samples of wood were taken from **22** trees and cut in the shape of a cube. The length of the edge of one of these cubes is **3.000** centimeters. This cube has a density of **0.250** grams per cubic centimeter. What is the mass of this cube, in grams?

Answer Preview:

Student-produced response directions

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Examples

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3.5	3.5 3.50 7/2	3 1/2 3 1/2

11



Mark for Review

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?

Answer Preview:



12



Mark for Review



The function f gives the product of a number, x , and a number that is 91 more than x . Which equation defines f ?

(A) $f(x) = x^2 + x + 91$

(B) $f(x) = x^2 + 91$

(C) $f(x) = x^2 + 91x$

(D) $f(x) = x^2 + 91x + 91$



13



Mark for Review



$$-2x^2 + 20x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

(A)

-68

(B)

-50

(C)

-32

(D)

0

Hide



Calculator



Reference



More

14



Mark for Review



$$y = 2x^2 - 15x + 23$$

$$y = x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

(A) -9 (B) -4 (C) 4 (D) 9

15



Mark for Review



The positive number a is 2,352% of the sum of the positive numbers b and c , and b is 84% of c . What percent of b is a ?

(A) 5,152%

(B) 2,436%

(C) 51.52%

(D) 24.36%





The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = 18(1.4)^{x+5}$

II. $g(x) = 18(1.16)(1.4)^{x+3}$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

17



Mark for Review

$$\frac{12x+28}{4} - \frac{s}{13} = r(x-8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s ?



The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

(A) 9

(B) $9\sqrt{2}$

(C) 18

(D) $18\sqrt{2}$



19



Mark for Review

A right triangular pyramid has exactly six edges. Each of these edges is 96 centimeters long. If the surface area of the pyramid is $k\sqrt{3}$ square centimeters, what is the value of k ?

20



Mark for Review



The cost of renting a piece of construction equipment for up to 5 days is \$270 for the first day and \$135 for each additional day. Which of the following equations gives the cost y , in dollars, of renting the equipment for x days, where x is a positive integer and $x \leq 5$?

(A) $y = 135x + 135$

(B) $y = 270x - 135$

(C) $y = 270x + 135$

(D) $y = 135x + 270$



A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

(A) $-14x + 1,540$

(B) $14x - 320$

(C) $-28x + 1,720$

(D) 360



22



Mark for Review



$$\begin{aligned}7x + 4y &= 3 \\ 35x + 20y &= 15\end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

(A) $\left(\frac{r}{5} + 3, -\frac{r}{5} + 15\right)$

(B) $\left(-\frac{4r}{7} + \frac{3}{7}, r\right)$

(C) $\left(r, \frac{7r}{4} + \frac{3}{4}\right)$



(D) $\left(r, -\frac{4r}{7} + \frac{3}{7}\right)$